Jing-Hua (Joseph) Chang

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Objective

Actively seeking a **full-time position** in hardware technology and engineering-related fields.

Education

University of California, San Diego (UCSD) *Sep.2025 - Dec.2026 (Anticipated)*

M.S. in Electrical and Computer Engineering | Track: Computer Engineering

- **Relevant coursework:** Principles in Computer Architecture (CSE 240A), Parallel Computer Architecture (CSE 240B), System Design for Machine Intelligence (CSE 240D), VLSI Integrated Circuits and Systems Design (ECE 260B), Low-power VLSI Implementation for Machine Learning (ECE 284), High-Speed Wireline Communication Circuits Systems (ECE 283 - Audit)

National Taiwan University (NTU) *Sep.2020 - Jun.2024*

B.S. in Engineering Science and Ocean Engineering | Track: Electronics and Information Engineering

- Ranked **2nd place** among 58 students in the 2023 academic year
- **Relevant coursework:** SoC Design Laboratory, Computer-aided VLSI System Design, Digital Signal Processing in VLSI Design

Skills

- **Coding:** RTL (Verilog, SystemVerilog), Python, C / C++, MATLAB, TCL scripting
- **Domain Knowledge:** Computer Architecture, VLSI System Design, Machine Learning, CDC/RDC, SerDes, Digital Design.
- **Tools:** Synopsys Design Compiler / Spyglass / Prime Time / Verdi, Cadence Virtuoso / Innovus / Xcelium, Xilinx Vivado, Intel Quartus, PyTorch, HSPICE.

Internships

Marvell, Digital IC Design Engineer - SerDes Team, Hsinchu *Jun.2025 - Aug.2025*

- Designed a high-speed (DDR) PRBS checker for a die-to-die SerDes test chip using Verilog, contributing to robust PHY functionality.
- Conduct CDC/RDC analysis, categorizing violation reports, identifying structural issues, and justifying safe waivers.
- Developed custom testbenches and automated test patterns to validate the PRBS checker, resolving functional bugs prior to verification.

Lumentum, Software Developer - Product Team, New Taipei *Nov.2023 - Jun.2024*

- Developed a portable software program for VCSELs die crack detection that expedited the testing process by 48x, and supported wafer mask generation & visualization for failure analysis. (OpenCV, Python, Docker)

Samsung Electronics, Field Application Engineer - SSD Memory Team, Taipei *July.2023 - Nov.2023*

- Designed an automation program streamlining the SSD FIO golden test on server systems, saving up to 72 hours of overhead. (Python)

Course Projects

- **RTL Development for Dual-Mode FlashAttention Accelerator:** Architected a unified GeneSys-based systolic and vector datapath in SystemVerilog to maximize MAC utilization during element-wise operations.
- **RTL Development for Low-Power CNN Accelerator:** Building the hardware acceleration system for ResNet and VGG algorithm with Verilog & PyTorch, synthesized using Intel Quartus on Cyclone FPGA. Achieve power reduction by 24%.
- **Physical Design for Transformer Accelerator:** Execute floorplanning, power grid, placement, clock tree synthesis (CTS), routing, and DRC/LVS using Cadence Innovus.

Research

System Energy Efficiency Lab *Jan.2026 - Mar.2026*

Graduate Research: Mapping-Agnostic Performance Bound for Heterogeneous Compute-in-Memory Architectures *Advisor: Tajana Rosing*

- Theoretical formulation of a fast and accurate performance bound for compute-in-memory architectures.

High Efficiency Circuit and System Laboratory *Jul.2022 - Feb.2024*

Undergraduate Research: MOS Current-Mode Logic Applied to VLSI Digital Design *Advisor: Jau-Horng, Chen*

- Designed & built the CML standard cell library (schematics/layout) on UMC 0.18 um CMOS 1.8V process using Cadence Virtuoso.